

# Restauração de imagens desfocadas (*Image deblurring*)

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11/02/2025

# Blurring



(a) Imagem original



(b) Imagem degradada por um *blurring*

- Matematicamente, o *blurring* pode ser representado por um operador:

$$\mathbf{y} = B(\mathbf{x}),$$

em que  $\mathbf{x}$  é a imagem original e  $\mathbf{y}$  é a imagem degradada por um *blurring*.

- *Deblurring*  $\equiv$  recuperar  $\mathbf{x}$ , dado  $\mathbf{y}$ :

$$\mathbf{x} = B^{-1}(\mathbf{y}).$$

- *Blurring* linear e invariante no espaço:

$$\mathbf{y} = \mathbf{k} * \mathbf{x}.$$

- *Deblurring*  $\equiv$  Deconvolução
- No domínio da frequência:

$$\mathcal{F}(\mathbf{y}) = \mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{x}).$$

# Uma primeira abordagem ingênua

- Filtragem inversa:

$$\mathcal{F}(\mathbf{x}) = \frac{\mathcal{F}(\mathbf{y})}{\mathcal{F}(\mathbf{k})}$$

- No espaço:

$$\hat{\mathbf{x}} = \mathcal{F}^{-1} \left( \frac{\mathcal{F}(\mathbf{y})}{\mathcal{F}(\mathbf{k})} \right)$$

- Em um cenário mais realista:

$$\mathbf{y} = \mathbf{k} * \mathbf{x} + \boldsymbol{\eta} \Leftrightarrow \mathcal{F}(\mathbf{y}) = \mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{x}) + \mathcal{F}(\boldsymbol{\eta}).$$

- Aplicando a filtragem inversa:

$$\hat{\mathbf{x}} = \mathcal{F}^{-1} \left( \frac{\mathcal{F}(\mathbf{y})}{\mathcal{F}(\mathbf{k})} - \frac{\mathcal{F}(\boldsymbol{\eta})}{\mathcal{F}(\mathbf{k})} \right)$$

- Amplificação de ruído

# Uma abordagem um pouco menos ingênua

- Filtro de Wiener:  $\hat{\mathbf{x}}$  é uma estimativa que minimiza

$$e^2 = \mathbb{E}|\mathbf{x} - \hat{\mathbf{x}}|^2.$$

- Resultado:

$$\mathcal{F}(\hat{\mathbf{x}}) = \left( \frac{1}{\mathcal{F}(\mathbf{k})} \frac{|\mathcal{F}(\mathbf{k})|^2}{|\mathcal{F}(\mathbf{k})|^2 + S_\eta/S_x} \right) \mathcal{F}(\mathbf{y})$$

# Deconvolução não cega vs Deconvolução cega

- Deconvolução não cega: Assume que o *kernel* que causou o blurring é conhecido.
- Deconvolução cega: O *kernel*,  $\mathbf{k}$ , precisa ser estimado junto com a imagem latente,  $\mathbf{x}$ .
  - ▶ Problema de inversão extremamente mal condicionado:
    - ★ Sistema indeterminado ( $\mathbf{y} = \mathbf{k} * \mathbf{x}$ )
    - ★ Não possui dependência contínua dos parâmetros.



# Solução via máximo *a posteriori* (MAP)

- A solução do máximo *a posteriori* é dada por

$$\hat{\mathbf{x}}, \hat{\mathbf{k}} = \arg \max_{\mathbf{x}, \mathbf{k}} P(\mathbf{x}, \mathbf{k} | \mathbf{y}).$$

- A probabilidade *a posteriori* é dada por

$$P(\mathbf{x}, \mathbf{k} | \mathbf{y}) = \frac{P(\mathbf{y} | \mathbf{x}, \mathbf{k}) P(\mathbf{x}) P(\mathbf{k})}{P(\mathbf{y})}$$

- Logo,

$$\begin{aligned} \hat{\mathbf{x}}, \hat{\mathbf{k}} &= \arg \max_{\mathbf{x}, \mathbf{k}} P(\mathbf{y} | \mathbf{x}, \mathbf{k}) P(\mathbf{x}) P(\mathbf{k}) \\ &= \arg \max_{\mathbf{x}, \mathbf{k}} \ln(P(\mathbf{y} | \mathbf{x}, \mathbf{k})) + \ln(P(\mathbf{x})) + \ln(P(\mathbf{k})). \end{aligned}$$

- Sabendo que  $\mathbf{y} = \mathbf{k} * \mathbf{x} + \boldsymbol{\eta}$  e assumindo que o ruído,  $\boldsymbol{\eta}$ , é gaussiano e de média nula, obtém-se:

$$P(\mathbf{y}|\mathbf{x}, \mathbf{k}) \propto e^{-\frac{1}{2}\|\mathbf{y}-\mathbf{k}*\mathbf{x}\|^2}$$

- Dessa forma, obtém se

$$\begin{aligned}\hat{\mathbf{x}}, \hat{\mathbf{k}} &= \arg \max_{\mathbf{x}, \mathbf{k}} -\frac{1}{2}\|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \ln(P(\mathbf{x})) + \ln(P(\mathbf{k})) \\ &\equiv \arg \min_{\mathbf{x}, \mathbf{k}} \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \lambda g(\mathbf{x}) + \gamma h(\mathbf{k}),\end{aligned}$$

em que  $g(\cdot)$  e  $h(\cdot)$  representam os conhecimentos *a priori* sobre as distribuições de  $\mathbf{x}$  e  $\mathbf{k}$ , respectivamente.

- Como escolher esses *a prioris*? E como estimar o máximo *a posteriori*?

# Deblurring Text Images via $L_0$ -Regularized Intensity and Gradient Prior

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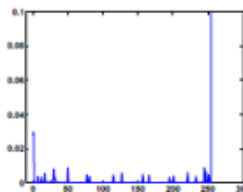
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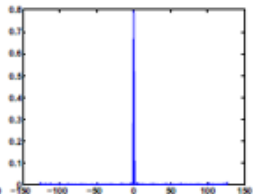
jspan@mail.dlut.edu.cn, zhu@ucmerced.edu, zxsu@dlut.edu.cn, mhyang@ucmerced.edu

**CVPR  
2014**

CVPR is the premier annual Computer Vision event comprising the main CVPR conference and several co-located workshops and short courses. With its high quality and low cost, it provides an exceptional value for students, academics and industry researchers.



(a)

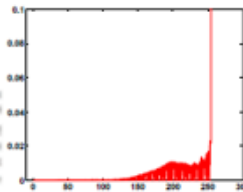


(b)

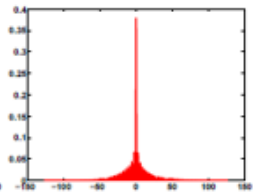
(c)

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(d)



(e)

(f)

## Uso da norma $L_0$ como *a priori*

- Definição da norma  $L_0$

$$\|\mathbf{v}\|_0 = \sum_i |v_i|^0$$

- Construção dos *a priori*s

$$g(\mathbf{x}) = \sigma \|\mathbf{x}\|_0 + \|\nabla_x \mathbf{x}\|_0 + \|\nabla_y \mathbf{x}\|_0$$

$$h(\mathbf{k}) = \|\mathbf{k}\|_2^2$$

- Solução final

$$\hat{\mathbf{x}}, \hat{\mathbf{k}} = \arg \min_{\mathbf{x}, \mathbf{k}} \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \lambda(\sigma \|\mathbf{x}\|_0 + \|\nabla_{\mathbf{x}} \mathbf{x}\|_0 + \|\nabla_{\mathbf{y}} \mathbf{x}\|_0) + \gamma \|\mathbf{k}\|^2$$

- Norma  $L_0$  não é diferenciável, e nem convexa.
- Como estimar  $\hat{\mathbf{x}}$  e  $\hat{\mathbf{k}}$ ?

- Processo iterativo

- ▶ Fixa-se  $\hat{\mathbf{k}}$  e otimiza-se  $\hat{\mathbf{x}}$ :

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \lambda(\sigma \|\mathbf{x}\|_0 + \|\nabla_x \mathbf{x}\|_0 + \|\nabla_y \mathbf{x}\|_0)$$

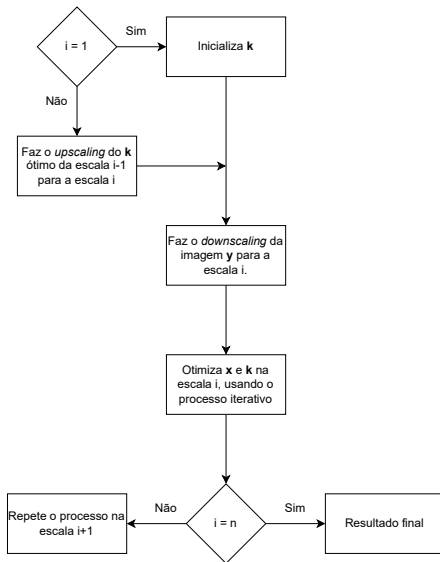
- ▶ Fixa-se  $\hat{\mathbf{x}}$  e otimiza-se  $\hat{\mathbf{k}}$ :

$$\hat{\mathbf{k}} = \arg \min_{\mathbf{k}} \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \gamma \|\mathbf{k}\|^2$$

- ▶ Os dois subproblemas são resolvidos alternativamente.

- Processamento multiescala.

# Processamento multiescala

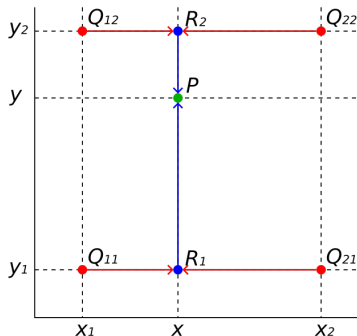




## Downscaling da imagem $y$

- Transformação *gamma*:  $y_{\text{out}} = A \cdot (y_{\text{in}})^\gamma$
- Filtro *anti-aliasing* (gaussiano)
- Interpolação bilinear

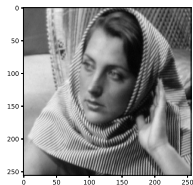
# Interpolação bilinear



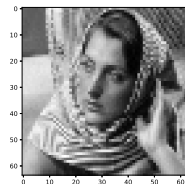
$$f(x, y_1) = \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21}),$$

$$f(x, y_2) = \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22}).$$

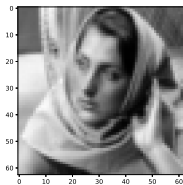
$$f(x, y) = \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2)$$



(a) Barbara  $256 \times 256$



(b) Barbara  $64 \times 64$ , com *aliasing*



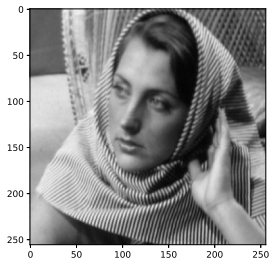
(c) Barbara  $64 \times 64$ , sem *aliasing*

## Upscaling do kernel, $\mathbf{k}$

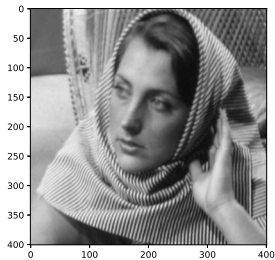
- Interpolação bicúbica

$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j.$$

- Leva em conta informação  $f, \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial x \partial y}$ . Cada um desses gera 4 equações (ou 4 condições) e os 16 coeficientes do polinômio são ajustados para satisfazê-las.



(a) Barbara  $256 \times 256$



(b) Barbara  $401 \times 401$

## Estimando $\mathbf{x}$ com $\mathbf{k}$

- $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} = \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \lambda(\sigma\|\mathbf{x}\|_0 + \|\nabla_x \mathbf{x}\|_0 + \|\nabla_y \mathbf{x}\|_0)$
- Norma  $L_0$  não é diferenciável.
- Introduz-se três variáveis auxiliares,  $\mathbf{u}$ ,  $\mathbf{g}_x$  e  $\mathbf{g}_y$ , correspondentes a  $\mathbf{x}$ ,  $\nabla_x \mathbf{x}$  e  $\nabla_y \mathbf{x}$ , respectivamente.
- $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} = \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \lambda(\sigma\|\mathbf{u}\|_0 + \|\mathbf{g}_x\|_0 + \|\mathbf{g}_y\|_0) + \beta\|\mathbf{x} - \mathbf{u}\|^2 + \mu(\|\nabla_x \mathbf{x} - \mathbf{g}_x\|^2 + \|\nabla_y \mathbf{x} - \mathbf{g}_y\|^2)$

## Estimando $\mathbf{x}$ com $\mathbf{u}$ , $\mathbf{g}_x$ e $\mathbf{g}_y$

- $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \beta \|\mathbf{x} - \mathbf{u}\|^2 + \mu (\|\nabla_x \mathbf{x} - \mathbf{g}_x\|^2 + \|\nabla_y \mathbf{x} - \mathbf{g}_y\|^2)$
- Expressão contém apenas termos diferenciáveis, e portanto esse subproblema possui uma solução analítica:

$$\hat{\mathbf{x}} = \mathcal{F}^{-1} \left( \frac{\overline{\mathcal{F}(\mathbf{k})} \mathcal{F}(\mathbf{y}) + \beta \mathcal{F}(\mathbf{u}) + \mu F_G}{\overline{\mathcal{F}(\mathbf{k})} \mathcal{F}(\mathbf{k}) + \beta + \mu (\overline{\mathcal{F}(\nabla_x)} \mathcal{F}(\nabla_x) + \overline{\mathcal{F}(\nabla_y)} \mathcal{F}(\nabla_y))} \right),$$

em que  $F_G = \overline{\mathcal{F}(\nabla_x)} \mathcal{F}(\mathbf{g}_x) + \overline{\mathcal{F}(\nabla_y)} \mathcal{F}(\mathbf{g}_y)$ .

- Derivação no apêndice

## Estimando $\mathbf{u}$ , $\mathbf{g}_x$ e $\mathbf{g}_y$ com $\mathbf{x}$

$$u = \begin{cases} x, & \text{se } |x|^2 \geq \frac{\lambda\sigma}{\beta}, \\ 0, & \text{caso contrário.} \end{cases}$$

$$g_x = \begin{cases} \nabla_x \mathbf{x}, & \text{se } |\nabla_x \mathbf{x}|^2 \geq \frac{\lambda}{\beta}, \\ 0, & \text{caso contrário.} \end{cases}$$

$$g_y = \begin{cases} \nabla_y \mathbf{x}, & \text{se } |\nabla_y \mathbf{x}|^2 \geq \frac{\lambda}{\beta}, \\ 0, & \text{caso contrário.} \end{cases}$$



# Estimando $\mathbf{x}$ com $\mathbf{k}$

- Novamente, a solução para  $\hat{\mathbf{x}}$  é obtida em um processo iterativo, em que fixa-se  $\mathbf{u}$ ,  $\mathbf{g}_x$  e  $\mathbf{g}_y$  para se obter  $\mathbf{x}$  e vice-versa.

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**Algorithm 1** Solving (6)

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**Input:** Blur image  $y$  and blur kernel  $k$ .

$x \leftarrow y, \beta \leftarrow 2\lambda\sigma$ .

**repeat**

    solve for  $u$  using (10).

$\mu \leftarrow 2\lambda$ .

**repeat**

        solve for  $g$  using (11).

        solve for  $x$  using (8).

$\mu \leftarrow 2\mu$ .

**until**  $\mu > \mu_{\max}$

$\beta \leftarrow 2\beta$ .

**until**  $\beta > \beta_{\max}$

**Output:** Intermediate latent image  $x$ .

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## Estimando $\mathbf{k}$ com $\mathbf{x}$

- $\hat{\mathbf{k}} = \arg \min_k \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \gamma \|\mathbf{k}\|^2$
- No espaço dos gradientes:  
 $\hat{\mathbf{k}} = \arg \min_k \|\nabla_x \mathbf{y} - \mathbf{k} * \nabla_x \mathbf{x}\|^2 + \|\nabla_y \mathbf{y} - \mathbf{k} * \nabla_y \mathbf{x}\|^2 + \gamma \|\mathbf{k}\|^2$
- Solução fechada:

$$\hat{\mathbf{k}} = \mathcal{F}^{-1} \left( \frac{\mathcal{F}(\nabla_x \mathbf{y}) \overline{\mathcal{F}(\nabla_x \mathbf{x})} + \mathcal{F}(\nabla_y \mathbf{y}) \overline{\mathcal{F}(\nabla_y \mathbf{x})}}{\overline{\mathcal{F}(\nabla_x \mathbf{x})} \mathcal{F}(\nabla_x \mathbf{x}) + \overline{\mathcal{F}(\nabla_y \mathbf{x})} \mathcal{F}(\nabla_y \mathbf{x}) + \gamma} \right)$$

- Derivação no apêndice

# Problemas de fronteira

- No domínio discreto, produto das fft's (na frequência)  $\equiv$  convolução circular.
- Logo, o produto  $\mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{x})$  é o equivalente, na frequência, da convolução ao longo de um período entre

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \vdots & & & \\ \vdots & & & \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} \text{ e } \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \vdots & & & \\ \vdots & & & \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

- Solução: *padding* (preenchimento).

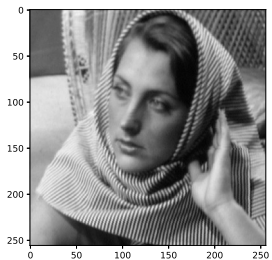
$$\begin{bmatrix} \mathbf{x} & \mathbf{0} \\ \mathbf{0}^t & \mathbf{0} \end{bmatrix} \text{ e } \begin{bmatrix} \mathbf{k} & \mathbf{0} \\ \mathbf{0}^t & \mathbf{0} \end{bmatrix}$$

- Na imagem  $\mathbf{x}$  preencher com zeros pode gerar artefatos indesejados devido à descontinuidade imposta. Uma solução mais interessante é crescer as dimensões da imagem de forma suave.

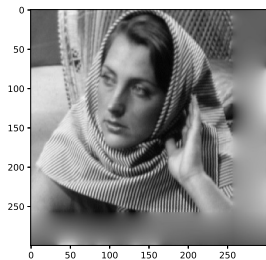
# REDUCING BOUNDARY ARTIFACTS IN IMAGE DECONVOLUTION

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(a) Barbara  $256 \times 256$



(b) Barbara  $300 \times 300$

## Centro do *kernel*

$$\begin{aligned} & \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 & 0 & 0 \\ 4 & 5 & 6 & 0 & 0 \\ 7 & 8 & 9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 4 & 5 & 6 & 0 & 0 \\ 7 & 8 & 9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & 6 & 0 & 0 & 4 \\ 8 & 9 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

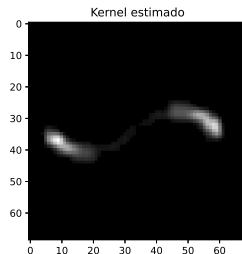
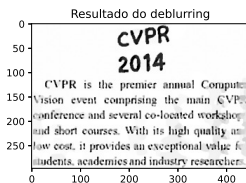
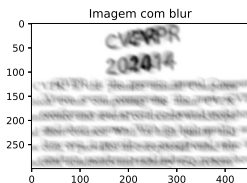
## Centro do *kernel*

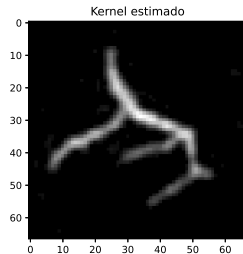
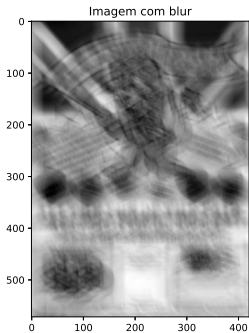
- Após a estimação do *kernel* (segundo subproblema), também é necessário corrigir o seu centro.
- Para isso, estima-se o centroide efetivo:

$$\bar{x} = \sum_i \sum_j j \mathbf{k}(i, j), \quad \bar{y} = \sum_i \sum_j i \mathbf{k}(i, j).$$

- Após isso, aplica-se uma operação de translação que faça com que o pixel  $(\bar{x}, \bar{y})$  seja movido para o centro do *kernel*.







## Resultado final pan



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# Fast Image Deconvolution using Hyper-Laplacian Priors

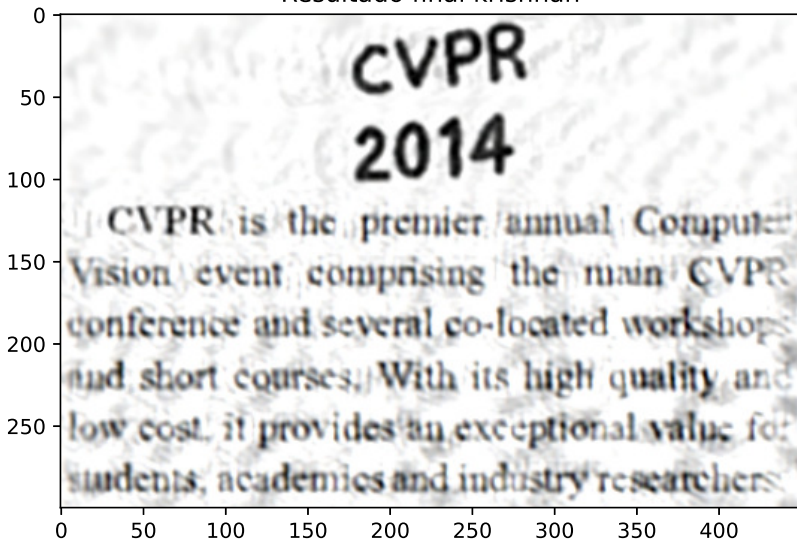
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- Algoritmo de deconvolução não cega
- Preserva detalhes
- Produz artefatos

## Resultado final krishnan



## Resultado final krishnan



# Filtro bilateral

$$I^{\text{filtered}}(x) = \frac{1}{W_p} \sum_{x_i \in \Omega} I(x_i) f_r(\|I(x_i) - I(x)\|) g_s(\|x_i - x\|),$$

- Filtra ruído, enquanto preserva bordas.
- O primeiro resultado apresentado pode ser aprimorado:
  - ▶ Subtrai o resultado obtido pelo algoritmo do Pan do resultado obtido pelo algoritmo do Krishnan.
  - ▶ Aplica o filtro bilateral a essa diferença.
  - ▶ Subtrai do resultado do algoritmo do Krishnan a diferença filtrada.



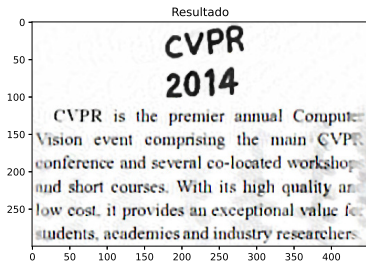
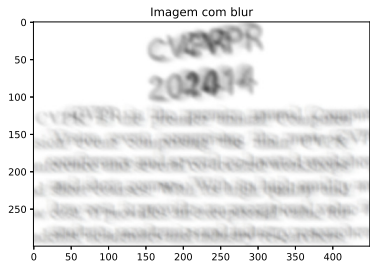
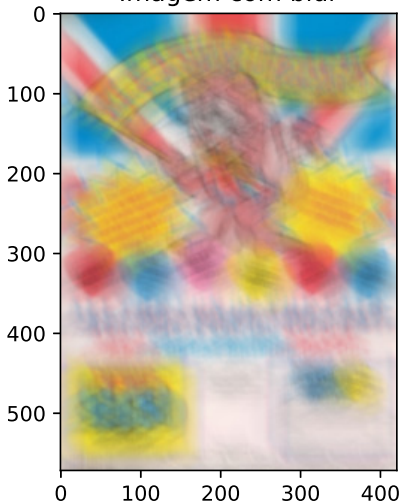


Imagem com blur



Resultado



OBRIGADO!

# Apêndice I: Solução fechada para estimação de $\mathbf{x}$ com $\mathbf{u}$ , $\mathbf{g}_x$ e $\mathbf{g}_y$

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} = \|\mathbf{y} - \mathbf{k} * \mathbf{x}\|^2 + \beta \|\mathbf{x} - \mathbf{u}\|^2 + \mu (\|\nabla_x \mathbf{x} - \mathbf{g}_x\|^2 + \|\nabla_y \mathbf{x} - \mathbf{g}_y\|^2)$$

Passando para o domínio da frequência:

$$\hat{\mathbf{x}} = \mathcal{F}^{-1}(\arg \min_{\mathcal{F}(\mathbf{x})} \|\mathcal{F}(\mathbf{y}) - \mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{x})\|^2 + \beta \|\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{u})\|^2 + \mu (\|\mathcal{F}(\nabla_x)\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{g}_x)\|^2 + \|\mathcal{F}(\nabla_y)\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{g}_y)\|^2))$$

Os seguintes resultados serão usados:

$$\textcircled{1} \quad \frac{\partial(\|\mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{y})\|^2)}{\partial \mathcal{F}(\mathbf{x})} = 2\mathcal{F}(\mathbf{k})(\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{y})})$$

$$\textcircled{2} \quad \frac{\partial(\beta \|\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{u})\|^2)}{\partial \mathcal{F}(\mathbf{x})} = 2\beta(\overline{\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{u})})$$

$$\textcircled{3} \quad \frac{\partial(\mu(\|\mathcal{F}(\nabla_x)\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{g}_x)\|^2 + \|\mathcal{F}(\nabla_y)\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{g}_y)\|^2))}{\partial \mathcal{F}(\mathbf{x})} =$$
$$2\mu(\mathcal{F}(\nabla_x)(\overline{\mathcal{F}(\nabla_x)\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{g}_x)}) + \mathcal{F}(\nabla_y)(\overline{\mathcal{F}(\nabla_y)\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{g}_y)}))$$

Logo, diferenciando toda a expressão e igualando a zero obtemos:

$$\begin{aligned}
 & -\mathcal{F}(\mathbf{k})\overline{\mathcal{F}(\mathbf{y})} + \mathcal{F}(\mathbf{k})\overline{\mathcal{F}(\mathbf{k})}\mathcal{F}(\mathbf{x}) + \beta\overline{\mathcal{F}(\mathbf{x})} - \beta\overline{\mathcal{F}(\mathbf{u})} + \\
 & \mu\mathcal{F}(\nabla_x)(\overline{\mathcal{F}(\nabla_x)\mathcal{F}(\mathbf{x})} - \mu\overline{\mathcal{F}(\mathbf{g}_x)} + \mu\mathcal{F}(\nabla_y)(\overline{\mathcal{F}(\nabla_y)\mathcal{F}(\mathbf{x})} - \mu\overline{\mathcal{F}(\mathbf{g}_y)}) = 0 \\
 \Rightarrow & \overline{\mathcal{F}(\mathbf{x})}(\mathcal{F}(\mathbf{k})\overline{\mathcal{F}(\mathbf{k})} + \beta + \mu(\mathcal{F}(\nabla_x)(\overline{\mathcal{F}(\nabla_x)} + \mathcal{F}(\nabla_y)(\overline{\mathcal{F}(\nabla_y)}))) = \\
 & \mathcal{F}(\mathbf{k})\overline{\mathcal{F}(\mathbf{y})} + \beta\overline{\mathcal{F}(\mathbf{u})} + \mu(\mathcal{F}(\nabla_x)\overline{\mathcal{F}(\mathbf{g}_x)} + \mathcal{F}(\nabla_y)\overline{\mathcal{F}(\mathbf{g}_y)}) \\
 \Rightarrow & \mathcal{F}(\mathbf{x}) = \frac{\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{y})} + \beta\overline{\mathcal{F}(\mathbf{u})} + \mu(\overline{\mathcal{F}(\nabla_x)\mathcal{F}(\mathbf{g}_x)} + \overline{\mathcal{F}(\nabla_y)\mathcal{F}(\mathbf{g}_y)})}{\mathcal{F}(\mathbf{k})\overline{\mathcal{F}(\mathbf{k})} + \beta + \mu(\mathcal{F}(\nabla_x)(\overline{\mathcal{F}(\nabla_x)} + \mathcal{F}(\nabla_y)(\overline{\mathcal{F}(\nabla_y)}))} \\
 \Rightarrow & \mathbf{x} = \mathcal{F}^{-1}\left(\frac{\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\mathbf{y})} + \beta\overline{\mathcal{F}(\mathbf{u})} + \mu(\overline{\mathcal{F}(\nabla_x)\mathcal{F}(\mathbf{g}_x)} + \overline{\mathcal{F}(\nabla_y)\mathcal{F}(\mathbf{g}_y)})}{\mathcal{F}(\mathbf{k})\overline{\mathcal{F}(\mathbf{k})} + \beta + \mu(\mathcal{F}(\nabla_x)(\overline{\mathcal{F}(\nabla_x)} + \mathcal{F}(\nabla_y)(\overline{\mathcal{F}(\nabla_y)}))}\right)
 \end{aligned}$$

## Apêndice II: Solução fechada para a estimação de $\mathbf{k}$ com $\mathbf{x}$

$$\hat{\mathbf{k}} = \arg \min_{\mathbf{k}} \|\nabla_x \mathbf{y} - \mathbf{k} * \nabla_x \mathbf{x}\|^2 + \|\nabla_y \mathbf{y} - \mathbf{k} * \nabla_y \mathbf{x}\|^2 + \gamma \|\mathbf{k}\|^2$$

Passando para o domínio da frequência:

$$\hat{\mathbf{k}} = \mathcal{F}^{-1}(\arg \min_{\mathbf{k}} \|\mathcal{F}(\nabla_x \mathbf{y}) - \mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_x \mathbf{x})\|^2 + \|\mathcal{F}(\nabla_y \mathbf{y}) - \mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_y \mathbf{x})\|^2 + \gamma \|\mathcal{F}(\mathbf{k})\|^2)$$

Os seguintes resultados serão usados:

- ①  $\frac{\partial(\|\mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_x \mathbf{x}) - \mathcal{F}(\nabla_x \mathbf{y})\|^2)}{\partial \mathcal{F}(\mathbf{k})} = \mathcal{F}(\nabla_x \mathbf{x})(\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_x \mathbf{x}) - \mathcal{F}(\nabla_x \mathbf{y})})$
- ②  $\frac{\partial(\|\mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_y \mathbf{x}) - \mathcal{F}(\nabla_y \mathbf{y})\|^2)}{\partial \mathcal{F}(\mathbf{k})} = \mathcal{F}(\nabla_y \mathbf{x})(\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_y \mathbf{x}) - \mathcal{F}(\nabla_y \mathbf{y})})$
- ③  $\frac{\partial(\gamma \|\mathcal{F}(\mathbf{k})\|^2)}{\partial \mathcal{F}(\mathbf{k})} = \gamma \overline{\mathcal{F}(\mathbf{k})}$

Logo, diferenciando toda a expressão e igualando a zero, temos:

$$\mathcal{F}(\nabla_x \mathbf{x})(\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_x \mathbf{x})} - \mathcal{F}(\nabla_x \mathbf{x})\overline{\mathcal{F}(\nabla_x \mathbf{y})} + \mathcal{F}(\nabla_y \mathbf{x})(\overline{\mathcal{F}(\mathbf{k})\mathcal{F}(\nabla_y \mathbf{x})} - \mathcal{F}(\nabla_y \mathbf{x})\overline{\mathcal{F}(\nabla_y \mathbf{y})} + \gamma\overline{\mathcal{F}(\mathbf{k})}) = 0$$

$$\Rightarrow \overline{\mathcal{F}(\mathbf{k})}(\mathcal{F}(\nabla_x \mathbf{x})\overline{\mathcal{F}(\nabla_x \mathbf{x})} + \mathcal{F}(\nabla_y \mathbf{x})\overline{\mathcal{F}(\nabla_y \mathbf{x})} + \gamma) = \mathcal{F}(\nabla_x \mathbf{x})\overline{\mathcal{F}(\nabla_x \mathbf{y})} + \mathcal{F}(\nabla_y \mathbf{x})\overline{\mathcal{F}(\nabla_y \mathbf{y})}$$

$$\Rightarrow \mathcal{F}(\mathbf{k}) = \frac{\overline{\mathcal{F}(\nabla_x \mathbf{x})}\mathcal{F}(\nabla_x \mathbf{y}) + \overline{\mathcal{F}(\nabla_y \mathbf{x})}\mathcal{F}(\nabla_y \mathbf{y})}{\mathcal{F}(\nabla_x \mathbf{x})\overline{\mathcal{F}(\nabla_x \mathbf{x})} + \mathcal{F}(\nabla_y \mathbf{x})\overline{\mathcal{F}(\nabla_y \mathbf{x})} + \gamma}$$

$$\Rightarrow \mathbf{k} = \mathcal{F}^{-1} \left( \frac{\overline{\mathcal{F}(\nabla_x \mathbf{x})}\mathcal{F}(\nabla_x \mathbf{y}) + \overline{\mathcal{F}(\nabla_y \mathbf{x})}\mathcal{F}(\nabla_y \mathbf{y})}{\mathcal{F}(\nabla_x \mathbf{x})\overline{\mathcal{F}(\nabla_x \mathbf{x})} + \mathcal{F}(\nabla_y \mathbf{x})\overline{\mathcal{F}(\nabla_y \mathbf{x})} + \gamma} \right)$$